COMPUTER ARCHITECTURE

LAB PROJECT PROPOSAL

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Project Title:

Adder and Subtractor Circuit Simulator

Submitted By

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Section	B
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Submitted To

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Project Introduction

Adder and Subtractor Circuit Simulator is an online application that simulates basic combinational circuits. The simulator will enable students to learn how binary numbers are added and subtracted through logic gates. The users can modify the input values and see the output of the various circuits.

Project Features

These are the four features of the application:

1). Half Adder and Full Adder: Users can alter input switches and see the output for Sum and Carry.

2). Half Subtractor and Full Subtractor: Users can alter input switches and see the output for Difference and Borrow.

3). Truth Tables: The truth tables for all circuits will be shown, and the present input combination will be highlighted.

4). Logic Gate Circuit Diagrams: Users will be able to see the logic gate circuit diagrams for all the circuits.

Division of Responsibilities

Member 1: Muhammad Afaq (Front End)

- Design the webpage using HTML and CSS.
- Create four tabs for the circuits.
- Design input switches and output indicators.
- Display circuit diagrams and truth tables.

Member 2: Muhammad Waleed Khan (Back End)

- Develop JavaScript functions for all four circuits.
- Implement Sum, Carry, Difference and Borrow calculations.
- Generate truth tables.
- Prepare 16 test cases to verify the results.

Technologies

- HTML
- CSS
- JavaScript

This application will operate through the web browser itself without the need for any database or server.

Expected Outcome

The final application will provide an interactive demonstration of half adder, full adder, half subtractor and full subtractor circuits. It will help us understand their input, outputs, truth tables and logic gate operations.